Double-entry bookkeeping for Social and Environmental accounting:

a framework to apply lessons from financial accounting to non-financial domains

WORKING PAPER VERSION: 2024, May 22

Sander Renes[1] and Jilde Garst[2]

[1] Delft University of Technology, corresponding author: s.renes-1@tudelft.nl, orcid-id: 0000-0003-2495-9219.
[2] Wageningen University, orcid-id: 0000-0003-3609-6078

Abstract:

To address both the marginalization of social and environmental data in accounting practice and the separation between the academic discourses on financial accounting and Social and Environmental Accounting, this paper analyses the underlying assumptions of the double-entry bookkeeping system and their relation to Social and Environmental Impact data. Our analysis identifies five axioms that shape double-entry bookkeeping and shows that the common valuation assumption in financial accounting is not required for double-entry bookkeeping. Adjusting this assumption allows the integration of Social and Environmental information in double-entry. This common framework bridges the divide between financial and non-financial information. It provides a common technical frame to better integrate existing knowledge in developing new accounting methods. We provide a proof of concept by applying our framework to company-level data on greenhouse gas emissions to show how the evaluation of performance changes for companies if we combine multiple performance dimensions into a single, consistent bottom-line number that generalizes net income.

Keywords:

bookkeeping system design, double-entry bookkeeping, ESG, sustainability reporting, social and environmental accounting, performance measurement

This paper has benefited greatly from discussions with Jeroen Suijs, Jeffrey Bouwer, Bauke Visser, and seminar audiences at the Erasmus University of Rotterdam and Erasmus Research Institute of Management.

Electronic copy available at: https://ssrn.com/abstract=4336278

1. Introduction

Double-entry bookkeeping is the foundation of our financial accounting system. However, the modern corporation is a much more complex organization than the banker-investor organizations of 16th-century Venice, where double-entry first came to prominence. These more complex organizations need to communicate with and depend on a much wider group of stakeholders than just the owners (Deegan, 2017; Stoelhorst & Vishwanathan, 2024). Reporting to a diverse audience will require more adaptability and customization of the reported numbers than is provided by current financial accounting practices (O'Dwyer & Unerman, 2016). To provide such flexibility, we conceptualize the double-entry bookkeeping system as a technical design that can be altered if required. Such alterations have occurred often during the historical developments of the bookkeeping system(Yamey, 1964). In our paper, we position bookkeeping as a tool that allows companies to provide relevant information to a wider group of decision-makers to broaden the scope of the reporting constituency. This is particularly relevant for social and environmental accounting (SEA), where societal actors have a clear need for reliable, verifiable, and auditable data (Hopwood et al., 2010).

Social and environmental accounting (SEA) has seen major developments in the last decades. These developments can be divided into three main fields that focus on how to: a) measure the environmental and social impact (SEI) of companies with generally accepted measures; b) record and utilize SEI data within companies; c) report SEI data to the external environment. Regarding the measurement, the list of SEI indicators has grown considerably (Christensen et al., 2021; O'Dwyer, 2011), with a recent inventory by the World Business Council of Sustainable Development identifying over 1400 SEI indicators in nine reporting standards (Astley et al., 2019)[1]. Consolidation efforts are on the way, both at the level of standards and regulations (CDP et al., 2020; EFRAG, 2021b, 2021a; European Commission, 2021; IFRS Foundation & Global Reporting Initiative, 2023; Science & 2023, n.d.; Taskforce on Nature-related Financial Disclosures, 2023; Uhrynuk et al., 2021; UNRISD, 2023; World Economic Forum, 2023), as well as at the organizational level (IFRS Foundation, 2023; The Value Reporting Foundation,

[1] The recently published draft of the ESRS standards that detail the reporting requirements of the CSRD indicate 10 SEI topics specified with 37 sub-topics and further specified with 73 sub-sub topics (ESRS1 AR16). These sub-sub-topics are not specified at the level of measurable indicators.

Electronic copy available at: https://ssrn.com/abstract=4336278

2021). In recording and using SEI information, many innovations are made in the area of information systems, especially concerning environmental management control systems (Guenther et al., 2016). Furthermore, the number of companies incorporating SEI-based targets in their executive remuneration policies has been increasing, leading to an increase in papers on the effectiveness of these incentives (Maas, 2018; Velte, 2020). In the field of sustainability reporting, while voluntary reporting was already increasing (KPMG, 2020), the EU has further accelerated this growth by mandating the reporting of SEI information for all companies with more than 500 employees (Directive 2022/2464/EU, 2022, art. 5).

However, in most of these developments, SEI information is still positioned as separate from financial information with separate measurement, recording, and reporting systems. Despite efforts of the former International Integrated Reporting Council (IIRC)[2] to integrate information on social and environmental impacts (SEI)[3] into financial reports, the majority of this information is still reported in separate sustainability reports (KPMG, 2020). In these reports, the connections between the financial, social, environmental, and broader economic impacts of a company are barely shown (Unerman et al., 2018). For recording, also separate information systems are developed for SEI information. While it is always possible to design new and separate systems for new types of information, this is often inefficient and slow. The design process can be accelerated and the likelihood of acceptance of the design increased if it is built on existing experience and knowledge (Houdet et al., 2020; Thornton, 2013). Developing accounting systems for SEI information that build on existing knowledge requires bridging this divide between financial and SEI systems and practices. As Hopwood et al. (2010, p.240) indicated, "integration of sustainability-related information within financial data collection processes results in data that is reliable, verifiable and auditable and helps to give the credibility needed to be taken seriously, both internally and externally."

[2] Now part of IFRS Foundation after their merger with SASB and consolidation with IFRS

[3] In this paper, data on social and environmental impacts (SEI data) refers to any data on a company's societal impacts that are not included in financial statements, thus including governance-related impacts and broader economic impacts. Other terms used in literature for this data are ESG data, sustainability data, non-financial data, environmental data, societal impact data, data on social value/capital, and data on environmental value/capital. While we understand that are important differences between these concepts, it is outside the scope of our paper to outline these differences.

Electronic copy available at: https://ssrn.com/abstract=4336278

In this paper, we address the system separation by developing a framework for double-entry bookkeeping and showing how it can be applied to both financial and social or environmental value flows of the company allowing the introduction of SEI into double-entry bookkeeping (Hopwood et al., 2010; Houdet et al., 2020). In section 2, we present a mathematical description of the double-entry bookkeeping system that does not rely on the assumption that all valuation is done for the owners. This structure allows us to think about SEI information in the same way as about financial information. Our framework highlights the key implications of accounting for a wider group of stakeholders: 1) the type of value flows can differ; 2) the need to be explicit about for whom the system is valuing; 3) the need to be explicit on the trade-offs made between the value generated for several stakeholders or to report (financial or societal) performance in several dimensions (D. P. Ellerman, 1986; Houdet et al., 2020). By making the valuations underlying a performance number explicit, we allow users to map the performance numbers from several Valuation assumptions, thus making the reports more user-friendly and customizable without losing the standardization required (Jia et al., 2022).

Having defined the framework and its implications, section 3 highlights how the design choices of double-entry systems can guide the application of decades' worth of accounting knowledge to the design of a unified bookkeeping system for integrating financial accounting and SEA. The comparison shows, however, that not all the accepted cannon within accounting can be applied one-on-one (Frame & O'Connor, 2011; Jones, 2010; Unerman et al., 2018). Section 3 thus provides an agenda for research in the technical aspects of SEA and several ethical considerations. Furthermore, to show how this double-entry unified bookkeeping system would work in practice, section 4 of our paper contains an empirical exercise. Using corporation-level greenhouse gas (GHG) emissions from the CDP (corporate data 2010-2020) and income statements of 5100 company-years, this section shows how income statements can be extended to unified performance statements. This section also highlights the importance of a unified performance measurement, the inclusion of one more dimension of performance already significantly affects the assessment of performance for the studied company-years. Depending on the sector between 10 and 30% of the observations go from net added value to net destroyed value.

Electronic copy available at: https://ssrn.com/abstract=4336278

By providing this new view on bookkeeping for SEI information – particularly, GHG information – our paper makes three contributions to accounting literature and practice. First, by reflecting on the possibilities and limitations of double-entry bookkeeping, we respond to the call of O'Dwyer & Unerman (2016, p.39) to develop "specific sets of knowledge on the substantive areas underpinning accounting for social sustainability" and work towards the technical preparedness for the required innovations.

Second, by having a closer look at the integration of SEI information in the current financial bookkeeping system, we provide a pathway for researchers and practitioners to apply the existing knowledge in accounting to investigate and experiment with unified reporting (Houdet et al., 2020). For practice, this unified reporting could prevent further marginalization of SEI information in the assessment of company performance (Humphrey et al., 2017; Zappettini & Unerman, 2016). For the academic community, this pathway can be the bridge between financial accounting and SEA that prevents isolated development of the fields (Cho, 2020).[4]

Third, our exploration of double-entry bookkeeping re-opens the academic discourse on our current bookkeeping system and accounting as an academic field. This discourse seems to have been dormant in the last decades and unable to stimulate innovation in practice and research (Demski, 2007; Fellingham, 2007). Using a technical design frame, our paper re-examines the fundamental assumptions of double-entry bookkeeping and thus evaluates the opportunities for innovation of the technical basis of accounting. This provides a fresh look, not just on what accounting is, but on what it can be. The current system of financial double-entry bookkeeping *assumes* it is for the use of the owner, investors, or other providers of financial resources.[5] We show this assumption is not required for double-entry accounting. Generalizing this assumption allows one to connect existing accounting knowledge to new Social and Environmental accounting issues on a solid theoretical basis. This provides an additional, timely, and relevant contribution of accounting research to practice (Rajgopal, 2021).

---

[4] Although in our framework we reference several discourses in social and environmental accounting, our paper is by no means a comprehensive overview of all developments in these fields, as this would be impossible to achieve in one paper. Cho (2020) provides a good start.

[5] See for, instance, Chapter 1, The Objective of General Purpose Financial Reporting, of FASB Concepts Statement No. 8, Conceptual Framework for Financial Reporting, or §1.2 IFRS Conceptual Framework for Financial Reporting 2018. For brevity, we will refer to these providers of financial resources as owners.

Electronic copy available at: https://ssrn.com/abstract=4336278

2. Mathematical framework of double-entry bookkeeping

Double-entry bookkeeping is designed, first and foremost, as a record-keeping system. Although double-entry was not designed in one step but rather evolved to meet the requirements of the time, we can work backward to show the underlying design choices. Section 2.1 proposes a framework for the current bookkeeping system in the form of five design choices (axioms) and several additional assumptions. By describing the design choices in the existing system, we aim to identify potential choices that can be adjusted to meet current and future needs to include SEI data. In Subsection 2.2, we adjust two of the axioms and the additional assumptions to create a bookkeeping system that allows the recording of SEI measures in the same system, it concludes by showing that the addition of SEI data for societal performance measurement does not weaken the system of double-entry bookkeeping. The development of this double-entry bookkeeping framework is complementary to the theoretical work on accounting valuation (Ijiri, 1967; Mattessich, 1964) and also draws heavily on earlier work on the mathematics of bookkeeping (D. Ellerman, 2011; D. P. Ellerman, 1986; Rambaud, 2010)

2.1 The five design choices of double-entry bookkeeping

The term 'organization' in accounting is defined through the entity assumption: an entity, sufficiently distinct from the rest of the world to have an identity. This identity entails that there are different aspects (assets, liabilities, etc.) of the entity that have some value to the stakeholders in the organization (see Mattessich (1964) for a formal definition). For our current purposes, we assume that the organization is a company with some assets and liabilities (aspects) whose values can be affected by economic events, or more formally:

Axiom 1 There exists some entity that contains *n* aspects with some value, with *n* accompanying accounts a_i to record the value of these aspects, with $1 \leq n < \infty$.

Identifying when and how the value of the entity changes, and thus what should be recorded, is an important step that is covered by materiality requirements and accounting standards. More formally this requires the identification of some set of events that alter the value of some aspect of the organization:

Axiom 2 There exists a measurable set of economic events Ω with elements ω^{τ} that change the value of an aspect *i* of an entity at moment τ.

Electronic copy available at: https://ssrn.com/abstract=4336278

If an economic event (e.g., a transaction) is identified, the next steps are to measure the size of the value change and update the books accordingly by changing the relevant numerical values. The accounting standards determine the way events are translated into numerical values. Formally, we define events as measurable sets, so the functions defined over those sets can be used to assign numerical values to each (combination of) events:

Axiom 3 The (changes) in aspect *i* are measured through real, signed measure $q_i(\cdot):\Omega \to R$ that maps the events to a value in real numbers.[6]

When all accounts are kept in the same units, moving value from one account to the next can be done by reducing the value in one element and increasing the value in another account by the same amount. Similarly, we can compare accounts directly. With physical units, this comparison is difficult since one has to answer a question like "Is 10 liters of oil more or less than 5 wheels?". Therefore, accounting usually makes the monetary unit assumption. This is a combination of two things; 1) the assumption of some common unit (Axiom 4), and 2) the Pure monetary unit assumption that states that this common unit is money. Because of this common, monetary unit, we can see the combined size of several inventories by simply adding up the numbers in the accounting system to see the total value of wheels and oil in inventory:

Axiom 4 - All accounts use the same unit: $\exists \mu \ s.t. \ \mu q_i = 1 \to a_i = 1 \simeq \ a_n = \ 1 \ \forall i,$

- This common unit (the unit used in account *n*) is a monetary unit (Pure monetary unit assumption)

where $\simeq$ denotes an equivalence relation to the user of the accounts.

A final property of our bookkeeping system is that every bookkeeping statement, whether it is a journal entry, a ledger, or a balance sheet, is balanced. Over the list, there is always as much debit value as there is credit value. This property is also described as (accounting) duality by Mattessich (1964) and Ijiri (1975). Ijiri (1975) discuss its origins either as a consequence of

[6] By requiring measurability, we are saying that there exists a σ-algebra, Σ, over Ω. The individual functions q_i can then be thought of as random processes defined over Ω.or Σ-measurable functions $q_i: \Omega \mapsto R$ (Williams, 1991). Under these assumptions, the numbers reported for each asset or liability in a company can be thought of as a time-series with an underlying chance process as we implicitly do when applying statistical analysis to the resulting financial accounts.

Electronic copy available at: https://ssrn.com/abstract=4336278

value definitions (causal duality) or as a consequence of descriptive reality (claims on assets and the assets should be worth the same). There is, however, another reason for duality in bookkeeping. We keep track of all the stocks and all the flows of value in the organization and do not allow for value to spontaneously appear or disappear. This makes bookkeeping for the values in the entity a closed system. The next Axiom perfectly captures this definition, since the restriction corresponds to the mathematical definition of a closed system:

Axiom 5 In any valid bookkeeping statement with n accounts, the sum of all values is 0:
$\sum_{i=1}^{n} q_i(\cdot) = 0 \, \forall \omega \in \Omega$

Under these axioms, we can assume that measures q_i take events as input and give a monetary value as output. The last question is then, value to whom?

Since its development by Venetian bankers who wanted to keep track of their foreign investments, accounting has developed as a tool for decision-making on behalf of the owners. To do so, double-entry bookkeeping systems has to provide periodic overviews of profit and a capital account for the owner (ten Have, 1973; Yamey, 2008). This function of double-entry bookkeeping let to a (often implicit) valuation assumption in accounting, all numbers are made consistent by measuring them relative to the financial value of the owner:

Valuation assumption A All value is relative to the financial claims of the owners.

By combining Valuation Assumption A and the axioms, we can single out the *n*-th account in the definition of Axiom 5 and rewrite the balance constraint as:

$$-q_n = \sum_{i}^{n-1} q_i$$

This is exactly how owners' equity is defined: whatever remains after subtracting the value of liabilities from the value of all assets. Furthermore, the change in owners' equity defines profits, which is used as the measure of the performance of the entity over a period. Comparing the value of owner's equity at moment *T* and moment *T*+1, we can rewrite profits as the change in all other accounts over the period:

Electronic copy available at: https://ssrn.com/abstract=4336278

$$-\left(q_n(\omega^{T+1}) - q_n(\omega^T)\right) = \sum_i^{n-1} q_i(\omega^T + 1) - \sum_i^{n-1} q_i(\omega^T) = \sum_i^{n-1} q_i(\omega^T + 1) - q_i(\omega^T)$$

Because we keep the bookkeeping system closed, the changes in the other $n - 1$ accounts determine the changes in the residual account and, thus, profits.

Lemma 1.

2. The capital account, owner's equity, closes the bookkeeping system by acting as a residual value account.
3. Performance, in the form of profit, is defined through the changes in all other accounts (assets and liabilities).

Note that in a single-entry system[7] that does not require balanced statements, all measures of value (q_i) – e.g., profit and owner's equity - can be independently set without a conflict of definitions. However, by closing the system in Axiom 5, the measure of owner's capital defines the profit measure: all value changes regardless of their relation to management's performance. If we define all measures to accurately capture the performance of the entity, then all assets and liabilities – i.e., owner's capital (q_n) – will only reflect performance-related value. Thus, defining the entity's value as a closed system creates an apparent conflict between income statement definitions and balance sheet definitions.

At this point, we have not discussed any of the normal accounting objects like debits, credits, ledgers, or journal entries. Appendix A.2 shows how these entities relate to the lists of n values that form an accounting statement in this section graphically and through the example of processing a journal entry.

[7] A note on definitions of single and double-entry. For mathematicians, double-entry implies that each element, so each account used, has two elements a debit and a credit element (one of which can be zero). For accounting scholars, double-entry usually refers to the fact that each entity, whether it is a journal entry or the entire ledger, requires at least 2 accounts with a positive value - one positive debit value and one positive credit value - to guarantee the entity is balanced. In this paper, we will use the later definition.

Electronic copy available at: https://ssrn.com/abstract=4336278

2.2 Generalizing double-entry bookkeeping

To record SEI measures in double-entry bookkeeping, we have to relax Axiom 4, the monetary unit assumption. When we relax the monetary unit assumption so that measures q_i can be in a non-monetary unit, we have to restate Axiom 4 and 5 :

Axiom 4* All values are linearly related: $\exists\, \lambda_i\ s.t.\ \lambda_i q_i = 1 \ \rightarrow a_i \simeq a_n = 1\ \forall i.$

where $\simeq$ denotes an equivalence relation to the user of the accounts.

Axiom 5* In any valid bookkeeping statement with *n* accounts, the λ-weighted sum of all values is 0: $\sum_i^n \lambda_i q_i = 0\ \forall \omega \in \Omega$

Axiom 4* implies that for any two elements *i,j* in our system, we can find a scalar $\mu_{I,j} = \frac{\lambda_i}{\lambda_j}$ such that 1 unit of q_i is equivalent in value to $\mu_{i,j}$ units of q_j. Note that any consistent unit (e.g., metric tons of CO2 equivalents) can be used for this equivalence, which relaxes the Monetary Unit Assumption. However, since we calculate economic values in monetary terms, in this paper, we will implicitly assume a monetary unit unless stated otherwise.

The difference between Axiom 4 and 4* is subtle. For physical goods, we could simply define q_i as the counted quantity, which would make λ_i the price per unit. Then, Axioms 4 and 4* have the same effect since the final value these measures result in is the same. However, while splitting the value in an account into quantities and prices makes interpretation easier, it is not always possible. For instance, you cannot count intangible assets and thus cannot define a physical measure to serve as q_i for intangible assets.

Axiom 5* requires the λ-weighted sum of the elements to be balanced. It would hold for vacuously if λ was all zeros, so at least two of the prices have to be non-zero.

Under Axiom 4*, every vector $\lambda \neq 0$ is a list of relative prices that defines a record-keeping system. The ratio between the two prices $\frac{\lambda_i}{\lambda_j}$ determines the rate at which units of account *i* are translated to units of account *j*. The prices in λ thus determine the trade-off between the different (changes in) assets and liabilities to determine performance and the value of the capital account. The set of all price lists Λ:$\{\lambda | \lambda \in R^n, \lambda \neq 0\}$ defines a family of record-keeping systems: one for

Electronic copy available at: https://ssrn.com/abstract=4336278

each list λ. In each, we can rewrite the balance constraint as: $\sum_i^{n-1} \lambda_i q_i = -\lambda_n q_n$, such that the crucial properties of double-entry bookkeeping, a performance and a residual capital account, are defined as well:

Proposition 1. A bookkeeping system defined by Axioms 1-3, 4*, and 5* is a double-entry bookkeeping system. All such systems have a residual value account that can be used as a capital account and admit a performance measure whose definition is a generalization of profit.

Proposition 2. All bookkeeping systems with *n* accounts that adhere to Axiom 1-3, 4*, and 5* can record the same states of the organization. There is a one-to-one relation between the states recorded in all such systems as long as all prices in all λ are non-zero.

Proof. See appendix.

Proposition 2 shows that we can change the λ_i without affecting the strength of our bookkeeping system. The only restriction is that we can guarantee the system is closed and thus meets Axiom 5*. However, with any additional value flow, the definition of the residual account automatically changes, and as a consequence, so does the performance measure (Valuation Assumption A and Lemma 1). A clear example is the addition of (net) greenhouse gas emissions of the entity. We measure the output q_{n+1} in metric tons (a real signed measure - Axiom 3) and the marginal value λ_{n+1} is defined through the marginal social cost of emissions. To keep the system closed with this additional value flow (Axiom 5*), we have to add the final value flow to the sum that defines the capital and performance measures: entity performance = profit - the social cost of emissions. However, this social cost violates Valuation assumption A, as its monetary value is not relative to the owners' financial claims but relative to the economic value owned by society. Thus, to accommodate this value flow in double-entry bookkeeping systems, we have to adopt a different valuation standard:

Valuation assumption A All value is relative to the valuation of some set of stakeholders to the entity.

Lemma 1'.

1. The capital account, in the form of stakeholders' equity, closes the bookkeeping system by acting as a residual value account.

Electronic copy available at: https://ssrn.com/abstract=4336278

4. Performance, in the form of value created for stakeholders, is defined through the changes in all other accounts (assets and liabilities).

Valuation assumption A' is one of the largest differences between SEA and current financial accounting. Although we can use the familiar frame of a double-entry system for SEA when accounting for externalities caused by the entity, we will have to shift our valuation assumptions to a wider definition of stakeholders and stakeholder values. Reporting for these new value flows – or indeed for different groups of stakeholders with different valuations and thus different lists of relative prices λ – can be done at a relatively low cost. For instance, when the GHG-corrected profit is the new, bottom-line-performance number reported by the annual statement, the owners can simply calculate profit as long as the underlying measures (GHG emissions) and values (marginal social cost of carbon) are reported. In our empirical test case, we show that this reporting causes no loss of information but can significantly change the assessment of performance.

3. Existing accounting research and social/environmental impact data, a research agenda

Although the axioms are a new way to present the structure of bookkeeping, they represent a centuries old bookkeeping system of which accounting scholars have debated every aspect. To tap into this large body of knowledge, we require clarity on what choices exist and what their consequences are. Using the axioms of our generalized bookkeeping system, we discuss these choices for SEI data and how the accounting profession has previously dealt with similar choices. In doing so, this section provides an agenda for future research in unified bookkeeping for social, environmental, and financial value.

3.1 What are the boundaries of the entity?

Axiom 1 puts two limits on what can be accounted for, which require making two design choices for an accounting system; 1) defining discrete boundaries between the entity and the rest of the world and what to do with externalities that cross these boundaries; 2) defining a finite number of elements in the ledger, i.e., a list of aspects to record and monitor.

Electronic copy available at: https://ssrn.com/abstract=4336278

Concerning the boundaries of the entity, many SEI topics represent externalities. These flows are difficult to reconcile with a narrow definition of the entity. For example, the Greenhouse Gas Protocol splits emissions into three scopes: direct emissions (Scope 1); indirect emissions related to energy consumption (Scope 2); and indirect emissions in the value chain for which the entity is responsible (Scope 3). The definition of the entity used in financial accounting would only account for direct emissions by the company (Scope 1) since all other scopes represent emissions produced by actors outside these boundaries. However, reporting only Scope 1 allows companies to 'outsource' their GHG emissions by simply buying the GHG-intensive activities from other entities, shifting emissions from Scope 1 to Scope 3 (Braam et al., 2016). In financial accounting, this scoping issue is a common problem for off-balance-sheet activities, leases, and pensions (e.g., Bauman, 2003; Spencer & Webb, 2015; Yu, 2013). Scandals like ENRON have forced regulators to increase reporting requirements to reduce problems, yet the control requirements inherent in the entity assumption have put limits on the inclusion of items on the balance sheet, pushing them to the notes instead. Similarly, one can envision solutions where scope 3 emissions are reported in notes rather than as part of the performance measure. An open question is whether that is sufficiently clear for users and sufficiently powerful for abatement incentives.

An alternative way to conceptualize this problem is to widen the definition of the entity to include the generation of energy (Scope 2), or even the entire value chain (Scope 3). When scope 3 emissions are brought under the entity definition, this creates accounting issues by including, for instance, energy companies in all clients' entity definitions. Solutions for such problems could be found in cost accounting methods. Assigning responsibility for emissions could be done in a similar fashion as assigning overhead costs or costs from staff departments over production departments. The lessons learned in developing cost accounting methods can thus be used to speed up the development of similar systems for SEA within the value chain entity (e.g. Kaplan & Ramanna, 2021). Furthermore, the division of responsibilities has direct consequences for the incentives for emissions abatement. Accounting research can provide insight into the pros and cons of various assignment methods, as well as how to best combine these methods with existing incentive systems (Kaplan & Norton, 2001a, 2001b).

Electronic copy available at: https://ssrn.com/abstract=4336278

Table 1. Design choices within double-entry systems

Axiom	Design choices	Examples of choice options
1. Entity	**Boundaries of the entity:** For which entity is the system keeping records?	**Entities:** Human society Nation Supply chain Organization Project or organization unit
2. Events & 3. Measurability	**Finite list of elements in the ledger:** Which elements to include? **Measurement system:** How to measure the impact of events?	**Element selection:** Financial materiality Impact materiality Double materiality **Measures:** Absolute quantity, signed measure-theoretic measures, the constant 1
4*. Unit Axiom	**Values and units:** What is the definition of value? Combine or separate all types of capital? What unit is used to compare accounts? If a monetary unit, which method is used for monetization?	**Value definition:** Financial capital Environmental capital Social capital **Units for comparison:** Physical units Monetary units Tokens **Monetization:** Value in use Historic value Willingness to pay/accept Abatement /avoidance cost Cost of damages Collective consent to pay
Valuation assumption & 5*. Balance Axiom	**Owner of the capital:** Measuring capital for whom? **Closing account:** What is (are) the closing account(s)?	**Capital owner:** The owner(s) A set of stakeholders Society in general **Closing account:** Owner's equity Societal Capital account Net planetary GHG emissions

3.2 What elements are material to record and how to measure them?

Axioms 2 and 3 define a measurement system: it puts a number to all SEI recorded. The measurable events ω, the shape of the measurement functions q_i, and their prices λ, are the core

Electronic copy available at: https://ssrn.com/abstract=4336278

of accounting standards and regulations. To make such standards useful for external comparisons, we need to ensure that measurements are both consistent over time and comparable between companies (Christensen et al., 2021; Cohen & Simnett, 2015; Jia et al., 2022). In other words, the relevant sets of events ω and functions q_i need to be defined for all entities and periods similarly.

In financial accounting, materiality places a limit on the number of reported items, limiting the relevant set of measures q_i. "Misstatements, including omissions, are considered to be material if they, individually or in the aggregate, could reasonably be expected to influence the economic decisions of users taken on the basis of the financial statements" IAASB (2021, p.332, ISA 320). These users are commonly assumed to be capital investors, so their response can be approximated through the response of stock markets (Ball & Brown, 1968, 2014; Clarkson et al., 2011). The EU's 'double materiality' definition clearly takes a wider perspective, indicating that SEI information is material when the information is "necessary for an understanding of the impact of the undertaking's activities on environmental, social and employee matters, respect for human rights, anti-corruption and bribery matters" (Directive 2022/2464/EU, 2022, par 29 of the preambule), and not just if it presents direct financial risks to the investors. However, the undetermined (relative) values of SEI cause issues (Garst et al., 2022). According to GRI and EFRAG, companies need to set quantitative thresholds to determine whether the SEI information represents a severe societal impact with a high likelihood of occurrence (Danes et al., 2023; GRI 3: Material Topics 2021, 2022). But can we say that the company's positive impact on 100,000 individuals is just as severe as its negative impact on 1,000 hectares of tropical forest? Without a common value to compare the impacts, how can these thresholds be set consistently to each other? If we cannot set consistent thresholds, how can companies determine whether an impact is material without making purely arbitrary choices?

Another issue with impact materiality based on impact is a chicken-egg problem: we have to measure an impact to know if it is material, while we only have to measure material impacts. There are two theoretical solutions to this issue. Either companies measure every impact and then determine which impact is material enough to report. This option runs into trouble quickly, as the list of things to measure is potentially endless (violating Axiom 1 and feasibility). Or standard-setters determine certain SEI are material by assumption and then mandate their

Electronic copy available at: https://ssrn.com/abstract=4336278

recording and reporting, as is done for GHG emissions and gender equality (Directive 2022/2464/EU, 2022; Directive - 2023/970, 2023). Defining a single, definitive list that covers all impacts for all companies is complex and can take quite some time, as shown by the development of the GRI Sector Program (*GRI - Sector Program*, n.d.) or the European sustainability reporting standards (*Sustainability Reporting Standards Interim Draft - EFRAG*, n.d.). More importantly, this list needs to be dynamic, as every day, new knowledge on grand challenges becomes available, and societal norms change (Arias et al., 2024; Eccles, 2020; Ferraro et al., 2015). Accounting scholars, in collaboration with other experts, could provide insights into how to develop such lists, taking into account the knowledge available and the changing societal norms (Bebbington & Unerman, 2018).

For SEI that need new measurements, inspiration can be drawn from the long list of measurement methods available in financial accounting. For instance, *IAS 16 - Property, Plant and Equipment* standardizes the carrying value (size of the stock) as well as the depreciation (flows) related to physical assets, defining the assets to be reported by all companies in every period. For complex and uncertain SEI, choosing a measurement system often comes down to choosing the preferred source of uncertainty. For example, the current method of recording GHG emissions is equivalent to expensing the social cost of carbon of CO2 while emitting. This requires a cost estimate equal to the lifetime damage of every marginal ton emitted. This treatment resembles the capitalization of long-term contracts by (pre-)booking the profits when the contract is signed, although in this case, the contracting party is society and the profits are losses. The long-term effects of GHG are uncertain, which implies that not only the length of the contract is uncertain but also the total damage caused during this period is (Rennert et al., 2021). An alternative method could be to calculate damages done by the stock of net GHG emissions of the entity over the past year. This method implies a different matching of costs to the activity, i.e., matching to the period in which the damage occurs instead of the period when the emission took place. This recording method treats the damages as the expense on a loan from society with variable interest. An (estimated) amount of GHG can be assigned to the entity based on past behavior, and then the entity can start (ac)counting from a given date. The cost per unit calculation only spans a year, which reduces the uncertainty around the cost estimate. However, in this method the estimation of the stock of past emissions is a new source of uncertainty. Each recording method thus places the uncertainty at a different position, changing the information

Electronic copy available at: https://ssrn.com/abstract=4336278

that can be obtained from both the unified income statement and the unified statement of financial position (Barker et al., 2020).

For some SEI, defining an appropriate measure for reporting might be complicated because the entity has no incentive to measure them. For GHG emissions, there is consensus that some emissions will occur in production (Gray, 2006), slave labor, however, is considered immoral and the only acceptable level is zero. Few companies would care to record the number of enslaved workers used in their production, while we would need such numbers to record their improvements. Accounting of the offending behavior is thus unlikely to work, which makes most incentive-based solutions untenable and leaves only bright-line rules and regulations (Caruana et al., 2021; Christ et al., 2019; Crane et al., 2022).

3.4 How to value and balance non-financial impacts?

Axiom 4* and Valuation Assumption A' together make it possible to translate the ratio scale measures q_i into a common unit. In this paper, we interpret the common unit used in the recording system as a monetary unit since it facilitates the comparison with traditional accounting methods, and we are used to considering corporate outputs in terms of monetary units. However, this is not necessary: as long as all values are expressed in terms of the Valuation of the same users, they can be directly compared in the different accounts (D. Ellerman, 2011; Houdet et al., 2020; Thornton, 2013).

To provide proper measures q_i and relative prices λ_i for SEI, we can rely on a large history of similar challenges over the development of double-entry accounting. In the mercantile trade, double-entry bookkeeping was first used to account for gold, commodities, and physical property, defined through physical counts (q_i) and historic prices (λ_i). Around the Industrial Revolution, bookkeepers needed to account for long-lived physical capital that deteriorated. Counts could still be used as q_i (e.g., number of machines), but the valuation (λ_i) was more difficult since the value of machines decreased over time, which led to the systems of depreciation. Similarly, intangible assets, such as patents and goodwill, presented an even larger challenge as neither measures (q_i) nor prices (λ_i) were available. Doubtful accounts, regulatory capital of banks, and new financial instruments all presented problems where prices have to be estimated based on models when markets do not exist. In this sense, adding social and

Electronic copy available at: https://ssrn.com/abstract=4336278

environmental value estimates to the accounting system is part of an ongoing process to expand the possibilities of bookkeeping as the primary technology of accounting, and we can take inspiration from these past innovations.

Much work is done outside of accounting on estimation methods and models that provide a financial equivalent to environmental impacts (Marrone et al., 2020). One idea receiving traction in accounting is the valuation of ecosystem services (Feger & Mermet, 2017; Ring et al., 2010; Sukhdev, 2009). Costanza et al. (2011) define ecosystem services as "the benefits people derive from functioning ecosystems" [p1]. By differentiating between ecological environments without human interaction and ecological environments that provide services for human users, the human user of ecosystem services will define what is to be counted (materiality, axiom 1) and how to value it (axiom 4). They differentiate between 'accounting' as an objective description of unit flows (measurement, our axiom 3) and 'assessment' as a more subjective exercise (valuation, our axiom 4) [p4].

Lessons learned from accounting can be combined with environmental economics to develop new governance systems for environmental impact. To illustrate, one of the challenges in ecosystem services valuation is the occurrence of 'bundled services': a river provides multiple services (drinking water, fishery grounds, transportation, etc.), which interact and thus lead to trade-offs (Feger & Mermet, 2017; Ring et al., 2010). In cost accounting, a similar issue is solved to assign overhead costs and costs of shared service centers either by defining the value of the whole cost via its constituting parts or by assigning the whole value to the different elements. Another example is the article by (Freeman & Groom, 2013) in which they apply insights into discount rates in cost accounting to long-term environmental impacts of projects.

While accounting for the social impacts of an entity has similar challenges for which existing accounting methods could provide answers, one challenge for valuation will not be solved with technological innovations. Putting a monetary value to an SEI means an entity can trade off the societal damage caused by the economic value created. Such trade-offs might not be ethically acceptable. To illustrate, while putting a monetary value on GHG emissions is commonly accepted, the same cannot be said for modern slavery. Economic trade-offs do not distinguish between the benefit of marginal reductions in slave labor and the cost of increased slave labor (Crane et al., 2022). Setting a price (i.e., how much is a free-life-year worth?) could be seen as

Electronic copy available at: https://ssrn.com/abstract=4336278

accepting the existence of enslaved laborers. However, a fundamental requirement for the integration into a double-entry accounting system is defining a marginal unit and a marginal price. As such, ethical concerns limit double-entry bookkeeping for SEI.

The final step for a unified double-entry bookkeeping system is to close the accounting system (Axiom 5*) and thus to define the stock of value 'invested in the company by nature or by society'. As Costanza et al. (2011, p.5) conclude: "Natural capital and ecosystem services are … changing the framing of the issue away from "jobs versus the environment" to a more balanced assessment of all the assets that contribute to human well-being and their interrelationships." Although scholars have developed macro-ecological balance sheets for national accounts (Feger & Mermet, 2017), at the company level, there is limited investigation into such balance sheets (Feger & Rambaud, 2020). In the next section, we show how a unified bookkeeping system can be closed through a simple example focusing on GHG emissions.

4. Empirical test case

In the previous section, we made explicit some of the underlying design choices that form our bookkeeping and accounting system. We used these design choices to identify where existing knowledge can or cannot be applied to new information demands for some problems in SEA. In this section, we go through a proof-of-concept exercise and add one example of SEI for which data is available to our bookkeeping system: GHG emissions that deteriorate the common environment. We explicitly go through the design choices to see how to make this flow fit in a consistent double-entry system.

4.1 Example Unified Performance Statement

To fix ideas, we first present a simplified Unified Performance Statement for the average company in our data. This example Unified Performance Statement starts with the standard financial reporting income statement ending in Net Income as the measure of financial performance for the owners. What follows is a selection of societal performance measures in SEI dimensions, the social cost of carbon in this example. Every part of the example Unified Performance Statement is reported in both their normal measure, as well as monetary units that can be used in the total performance measure.

Electronic copy available at: https://ssrn.com/abstract=4336278

Unified Performance Statement

		Measure (q)	positive value ($)	negative value ($)	(sub)totals ($)
FINANCIALS	Revenue		19,431.84		
	Expenses			17,176.74	
	Income before taxes				2,255.10
	Taxes			540.60	
	Net Income	1,714.50	1,714.50		1,714.50
SEI	GHG emissions (metr. Ton)	4,398,912			
	Social Cost of Carbon Emissions			226.96	
	Societal performance				-226.96
	SUMMARY				
	Unified Total Performance in dollar equivalent units				1487.54

Figure 1 Example of an Unified Performance Statement

Note that the income statement in its current form is still part of this performance statement. The prices in λ allow us to translate all other measures, like GHG emissions, to monetary values. To do the aggregation we assume that the relative price of profits is 1 by normalization. For each measure in the SEI section, we then define a monetary value that can be added to or subtracted from the rest of the performance measure.

Technically, the statement of the financial position could be similarly adjusted by adding separate capital accounts for SEI for each (the sum of SEI over time), what we refer to as societal performance. From a practical point of view, tracking the stocks in the statement of SEI-related value will be very difficult as most corporations have not recorded the history of their SEI, hence we will only focus on GHG for which some data now exists.

Electronic copy available at: https://ssrn.com/abstract=4336278

4.2 Adding SEI measures for societal performance to the double-entry system

To apply the double-entry system to unified performance as a combination of several performance measures, like net income and GHG emissions, we go through the choices identified by our axioms and assumptions.

Since we are including societal performance, taking the view of value relative to the financial value of the owners (Valuation assumption A) makes little sense. We, therefore, apply a value relative to the marginal value to society (Valuation assumption A'). Net income, as a measure of financial performance for the owner, already stems from a double-entry system and thus meets Axioms 1-5. The only remaining question is what value of λ to apply to translate the financial value to the owners to the value of its SEI. Since we can normalize the relative prices, we assume that the translation from profit to SEI-related value is done at a price of 1. GHG emissions are measured (q_i) in metric tons emitted. The value (λ_i) to society of an additional metric ton of CO2 emissions is given by the social cost of carbon. We follow the recent Brookings Institute report on CO2 prices (Rennert et al., 2021) and assume that this is equal to $56 per ton.

For GHG emissions, we focus on the organization's own emissions calculated as the total emissions in scope 1 (direct emissions), scope 2 (emissions caused by the energy consumed by the organization), and biogenic emissions (Biogenic, or biologically sequestered CO2). In this definition of GHG emissions by the company, we exclude scope 3 emissions (emissions in the value chain) for three reasons. First, they are not purely under the control of the focal company, hence claiming that the value flow affects some aspect of the company (Axioms 1 and 2) appears not to hold for this value flow. Second, since scope 3 covers the entire production chain from raw resources to final consumption for each good most emissions would be reported on by several companies (Peters, 2010), leading to undesirable double-counting. Third, their measurement is much less precise than that of the other two emission scopes, and fewer companies report on it, which decreases the reliability of this data.

GHG emissions of the company – in metric tons of CO2 equivalents – meet the requirements of measurement (Axiom 3). Through the social cost of carbon, we can linearly relate it to its relevant dollar value (Axiom 4*). To meet the final criterium (Axiom 5) and close the bookkeeping system, there has to be a capital account that keeps track of the total emissions and abatements done by the company in its history. The Social Cost of Carbon is a Cost to society,

Electronic copy available at: https://ssrn.com/abstract=4336278

this flow is expected to have a net Debit value. Its 'consumption by the entity' should thus be offset by a Credit-type capital account on the balance sheet. This environmental capital account tracks the total value subtracted from society via greenhouse gas emissions similar to the accounts keeping track of value received from owners (equity) and lenders (liabilities).

Double-entry bookkeeping does not prevent double counting of a value flow, something we deal with it every time financial statements are consolidated. The costs made by one company are the profits of another. So, in consolidated financial statements, we cancel out the within-group transactions. Similar problems occur in accounting for greenhouse gas emissions. The Scope 1 emissions of the energy sector are the Scope 2 emissions of their customers. Therefore, the energy sector is excluded from this proof-of-concept.

4.3 The Data

We obtain data on corporate GHG emissions through the Carbon Disclosure Project (CDP), a corporate survey on climate change 2010-2020 (fiscal years 2009-2019). These surveys provide, among a host of other topics, voluntarily reported emissions in metric tons $CO2$ equivalents, following the GHG Protocol Corporate Standard (https://ghgprotocol.org/).

We have data on direct emissions for 14594 company-years of 1692 unique companies. These are both locally and internationally operating companies, both private and publicly listed. We will assume that if either scope 1 or scope 2 emissions have no positive number reported, the reported number is 0. If all three reported emissions are 0, we assume no reporting occurred and these observations are removed from the analysis. We match the remaining observations with data obtained from the annual statements obtained from WRDS, North America, and WRDS, global. We match the datasets based on the fiscal year and either Ticker (tic), or ISIN. Then we remove companies for which either the Net Income (ni) or Total Tax (txt) variable is not available. To prevent double counting GHG emissions, we remove companies whose main business is electric power generation (NAICS codes 22111) from our data. This leaves us with 5100 company-years corresponding to 646 companies. Descriptive statistics of our final sample are shown in Table 2. We use the taxes paid to show how this (self-)selection of companies compares to other companies in the same period. We find that they have an effective tax rate of 0.2293477% which is on par for this time period (Dyreng et al., 2017). Even though these

Electronic copy available at: https://ssrn.com/abstract=4336278

companies choose to report GHG emissions, they seem to be relatively normal corporate citizens in this regard.

Table 2 Descriptive Statistics

		N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
	Revenue (revt)	5,100	19,431.84	37,191.98	-9	2,285.9	18,854	483,521
	Net income (ni)	5,100	1,714.50	4,610.03	-23,119	71.8	1,692.5	98,806
	Cost of emissions	5,100	226.96	783.62	0.0003	4.42	94.51	10,219.56
	Corporate income tax (txt)	5,100	540.60	1,939.09	-34,831	9.7	530.2	31,051
	Effective Tax Rate	5,100	0.23	1.62	-37.54	0.10	0.34	49.42
Location	Africa	5,100	0.04	0.20	0	0	0	1
	Americas	5,100	0.67	0.47	0	0	1	1
	Asia	5,100	0.02	0.15	0	0	0	1
	Europe	5,100	0.23	0.42	0	0	0	1
	Oceania	5,100	0.03	0.18	0	0	0	1
NACIS-2 Industry	Heavy industry	5,100	0.10	0.30	0	0	0	1
	Manufacturing	5,100	0.43	0.49	0	0	1	1
	Services	5,100	0.32	0.46	0	0	1	1
	Trade and logistics	5,100	0.11	0.31	0	0	0	1
	Other industries	5,100	0.05	0.22	0	0	0	1

4.4 Societal performance and financial performance compared

To show how much influence the reporting of SEI-related value flows has on the perceived performance of the companies in our data set, we compare the sizes of performance in our example dimensions (Net Income and SCC) in Figure 1. In each figure, loss years are in red, profitable years in blue and the diagonal black line shows where SCC and net income are equally large. We see a significant group of observations above the diagonal line, indicating that non-financial performance has a larger impact on society than financial performance for those corporate years. The reported performance of the heavy industry and manufacturing sectors is the most affected by SCC. What is noticeable, however, is a small number of companies in the

Electronic copy available at: https://ssrn.com/abstract=4336278

Services and Other sectors that have high profitability but score very negatively in terms of SCC. These companies might perform well for their owners, but are very costly for wider society.

Figure 1A



Figure 1B



Figure 2 Comparison SCC with Net Income

Electronic copy available at: https://ssrn.com/abstract=4336278

To show how much this adjustment already matters to bottom-line performance, Table 3 shows what share of the company-years ends below 0 if performance is measured as Net Icnome, or when performance is measured as Net Income minus the Social Cost of Carbon.[8]
The Heavy Industry sector scores the worst: in 27.41% of the years, they report a loss, while the social cost of carbon is larger than profit for more than half of the company-years. The Trade and Logistics sector consistently scored best, with only 6.41% of the observations reporting a loss, and 15.9% had higher SCC than profit. For the Manufacturing, Services, and 'Other' sector, the picture is fairly consistent, between 12.8 and 15.3 % of the company-years report a loss, and between 19.5-27.9% have a higher SCC than they have positive profit. Overall, for most sectors, the effect of adjusting for CO2 is significant. Many companies that have a positive performance in terms of the financial gain of owners have a negative performance after adjusting for the cost of their GHG emissions.

Table 3 Percentage of company-years with negative performance (i.e., Net Income – SCC)

Industry	Loss-years	NI-SCC negative
Heavy industry	27.41	52.51
Manufacturing	13.25	19.47
Trade and logistics	6.41	15.93
Services	15.26	23.64
Other	12.75	27.89

5. Discussion and conclusion

SEI data is often treated as separated from or fundamentally different from financial data (KPMG, 2020; Unerman, 2020; Unerman et al., 2018). We argue that this separation is not fundamental. On a conceptual level, both can be seen as measures of the organization's

[8]As a robustness check, we also split observations according to fiscal years and regions in the Appendices. We find that over time the SCC of the reporting companies decreases slightly, which is to be expected as emissions have become more salient over time, and through our data, we have preselected companies that are voluntarily measuring emissions. It is therefore very likely that these companies are reducing their emissions. We see no clear geographical patterns.

Electronic copy available at: https://ssrn.com/abstract=4336278

performance in creating or destroying and stewardship of resources we as humans value. We formalize this conceptual similarity by creating a framework for unified bookkeeping, combining financial data for financial performance and SEI data for societal performance. We present this framework through five technical properties that have to be met to qualify as a double-entry bookkeeping system. These properties are closely related to the definitions of the entity assumption, economic events, the common unit of value in accounting, and the closed-system assumption implied in accounting duality (Ijiri, 1975; Mattessich, 1964). Additionally, we show that valuation on behalf of financial investors, often implicit in accounting research and practice, is unnecessary for double-entry bookkeeping. Yet, this assumption determines reported numbers on a fundamental level. Simultaneously, we can show that translating from this valuation standard to any other sufficiently strong standard is possible without loss of information and within the double-entry bookkeeping framework.

Our common framework for both financial and SEI bookkeeping highlights the elements, problems, and potential solutions that are common to both. Our framework aims to bridge the divide between the financial accounting and SEA communities by providing a common vocabulary and an agenda for building a new generation of accounting systems.
The framework can help further the development of SEA scholars to more directly relate their insights to existing bodies of knowledge within financial and management accounting. The framework also presents an opportunity for traditional accounting researchers to re-examine the field's foundations and draw inspiration from SEA research.

The framework is only a start: our article does not provide solutions to all challenges in SEA nor is it a step-by-step “how-to”-guide for building a unified bookkeeping system. For accounting scholars, the framework could serve as an instrument to compare multiple recording and reporting systems that are suggested, developed, and offered in practice. For practitioners and standard-setters, the framework could serve as an instrument to stimulate reflection on choices that are made in developing and using accounting systems.

By positioning the technical aspects of the bookkeeping system as design choices, we show that the designs of our accounting systems depend on the choices we make as an accounting community. These choices depend on the audiences' information needs addressed by the systems and the norms of our academic and professional communities. Which information is valuable, if

Electronic copy available at: https://ssrn.com/abstract=4336278

and how to monetize it, and whether it can be compared to other valuable information and thus potentially traded off in decision-making are all determined as much by our accounting standards and systems as by our moral judgments. Questioning the current systems and experimenting with alternative (re)configurations, as we did in this article, is crucial for the accounting scholarly community to maintain its academic and practical relevance in the creation and use of information in decision-making (Demski, 2007; Fellingham, 2007; Rajgopal, 2021). It allows us to look not just at what accounting is, but what it can be, and thus allows the profession to respond to the information demands of the time, both now and in the future.

Our choices as accounting community have considerable consequences for how companies, markets, and our economic systems respond to grand challenges. We as academics should thus not make recommendations for these choices light-heartedly nor should we position these choices as absolute. There is not one “right choice”: all systems need to adapt to the needs of the context in which they are used and the preferences and norms of their users. As we show, a broader definition of the users than just the owner or financial capital provider is possible without violating double-entry requirements, and for social and environmental challenges, this may be necessary. Additionally, these users' information needs and norms will change over time (Arias et al., 2024; Eccles, 2020; Ferraro et al., 2015). Some of these changes can be responded to with slight changes in our accounting standards, but other changes might need reconsideration and redesign of the foundations of our accounting systems. Our framework will allow scholars and practitioners alike to navigate these changes by identifying which design choice needs to be reconsidered. Our technical choices for double-entry bookkeeping also show boundaries to using double-entry. Although these boundaries do not restrict the use of double-entry for SEI-related values, accounting through double-entry does imply strict measurability requirements, a clearly defined and closed entity, as well as well defined measurable events. Not all SEI criteria can meet these requirements at this point in time, whether they will ever is an open question.

With this said, we also need to reflect on our own position. Our effort to explicate the choices and assumptions in double-entry bookkeeping reflects our own norms and assumptions. We, therefore, welcome scholars and professionals to provide feedback on any choices or assumptions we might have missed or misinterpreted. Finally, our framework is not representative of all design choices in accounting. Our framework presents a stylized version of

Electronic copy available at: https://ssrn.com/abstract=4336278

double-entry bookkeeping. While bookkeeping is at the foundation of modern financial accounting, it is not the only possible system or system in use. Investigating other accounting systems might bring other (standardized) design choices and assumptions to the surface that hinder innovation in SEA. Furthermore, innovation in SEA will continue and our economic systems will transform. These changes can bring design choices for bookkeeping systems to the surface that we didn't consider. We welcome the accounting community to continue investigating new accounting systems to extend or revise our framework. More importantly, we invite scholars to continue the discourse on the choices and assumptions underlying the existing systems that provide the foundation of the accounting profession, even those in place since the 16th century.

References

Arias, D., Barriola, X., & Loza Adaui, C. R. (2024). Corporate purpose and early disaster response: Providing evidence of dynamic materiality? *Business Strategy and the Environment*. https://doi.org/10.1002/bse.3707

Astley, D., Beanland, A., Blower, L., Tahtinen, J., & Guthrie, J. (2019). Creating alignment and harmonization in the ESG reporting landscape: 30+10 places to start. In *EMAN Conference on Biodiversity and Natural Capital Accounting*. World Business Council for Sustainable Development.

Ball, R., & Brown, P. (1968). An Empirical Evaluation of Accounting Income Numbers. *Journal of Accounting Research*, *6*(2), 159. https://doi.org/10.2307/2490232

Ball, R., & Brown, P. R. (2014). Ball and Brown (1968): A Retrospective. *The Accounting Review*, *89*(1), 1–26. https://doi.org/10.2308/ACCR-50604

Barker, R., Ahmed, A., Barth, M., Clubb, C., Cooper, S., Dichev, I., Harris, T., Linsmeier, T., Mcgeachin, A., Pope, P., Rajgopal, S., Ryttersgaard, T., Schipper, K., Teixeira, A., & Whittington, G. (2020). Moving the Conceptual Framework Forward: Accounting for Uncertainty. *Contemporary Accounting Research*, *37*(1), 322–357. https://doi.org/10.1111/1911-3846.12585

Bauman, M. P. (2003). The Impact and Valuation of Off-Balance-Sheet Activities Concealed by Equity Method Accounting. *Accounting Horizons*, *17*(4), 303–314.

Electronic copy available at: https://ssrn.com/abstract=4336278

https://doi.org/10.2308/ACCH.2003.17.4.303

Bebbington, J., & Unerman, J. (2018). Achieving the United Nations Sustainable Development Goals: An enabling role for accounting research. *Accounting, Auditing and Accountability Journal*, *31*(1), 2–24. https://doi.org/10.1108/AAAJ-05-2017-2929

Braam, G. J. M., Uit De Weerd, L., Hauck, M., & Huijbregts, M. A. J. (2016). Determinants of corporate environmental reporting: the importance of environmental performance and assurance. *Journal of Cleaner Production*, *129*, 724–734. https://doi.org/10.1016/J.JCLEPRO.2016.03.039

Caruana, R., Crane, A., Gold, S., & LeBaron, G. (2021). Modern Slavery in Business: The Sad and Sorry State of a Non-Field. *Business and Society*, *60*(2), 251–287. https://doi.org/10.1177/0007650320930417

CDP. (2020). *CDP corporate data*. Https://Www.Cdp.Net/.

CDP, CDSB, GRI, IIRC, & SASB. (2020). *Statement of Intent to Work Together Towards Comprehensive Corporate Reporting*. *September 2020*.

Cho, C. H. (2020). CSR accounting "new wave" researchers: "step up to the plate"... or "stay out of the game." *Journal of Accounting and Management Information Systems*, *19*(4), 626–650. https://doi.org/10.24818/jamis.2020.04001

Christ, K. L., Rao, K. K., & Burritt, R. L. (2019). Accounting for modern slavery: an analysis of Australian listed company disclosures. *Accounting, Auditing and Accountability Journal*, *32*(3), 836–865. https://doi.org/10.1108/AAAJ-11-2017-3242

Christensen, H. B., Hail, L., & Leuz, C. (2021). Mandatory CSR and sustainability reporting: economic analysis and literature review. *Review of Accounting Studies 2021 26:3*, *26*(3), 1176–1248. https://doi.org/10.1007/S11142-021-09609-5

Clarkson, P., Hanna, J. D., Richardson, G. D., & Thompson, R. (2011). The impact of IFRS adoption on the value relevance of book value and earnings. *Journal of Contemporary Accounting & Economics*, *7*(1), 1–17. https://doi.org/10.1016/J.JCAE.2011.03.001

Cohen, J. R., & Simnett, R. (2015). CSR and Assurance Services: A Research Agenda. *AUDITING: A Journal of Practice & Theory*, *34*(1), 59–74. https://doi.org/10.2308/AJPT-

Electronic copy available at: https://ssrn.com/abstract=4336278

50876

Costanza, R., Kubiszewski, I., Ervin, D., Bluffstone, R., Boyd, J., Brown, D., Chang, H., Dujon, V., Granek, E., Polasky, S., Shandas, V., & Yeakley, A. (2011). Valuing ecological systems and services. *F1000 Biology Reports*, *3*(1). https://doi.org/10.3410/B3-14

Crane, A., LeBaron, G., Phung, K., Behbahani, L., & Allain, J. (2022). Confronting the Business Models of Modern Slavery. *Journal of Management Inquiry*, *31*(3), 264–285. https://doi.org/10.1177/1056492621994904

Danes, G. S., Heintges, S., Mazzeo, P., Villacorta, V., & Trouvain, E. (2023). Implementation guidance for the materiality assessment. In *Efrag*.

Deegan, C. (2017). Twenty five years of social and environmental accounting research within Critical Perspectives of Accounting: Hits, misses and ways forward. *Critical Perspectives on Accounting*, *43*, 65–87. https://doi.org/10.1016/J.CPA.2016.06.005

Demski, J. S. (2007). Is accounting an academic discipline? *Accounting Horizons*, *21*(2), 153–157. https://doi.org/10.2308/acch.2007.21.2.153

Directive 2022/2464/EU. (2022). *As regards corporate sustainability reporting*. *L 322*, 15–80. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022L2464

Dyreng, S. D., Hanlon, M., Maydew, E. L., & Thornock, J. R. (2017). Changes in corporate effective tax rates over the past 25 years. *Journal of Financial Economics*, *124*(3), 441–463. https://doi.org/https://doi.org/10.1016/j.jfineco.2017.04.001

Eccles, R. G. (2020). Dynamic Materiality And Core Materiality : A Primer For Companies And Investors. *Forbes*, 1–9. https://www.forbes.com/sites/bobeccles/2020/01/17/dynamic-materiality-and-core-materiality-a-primer-for-companies-and-investors/#1c23b4a42e6a

EFRAG. (2021a). *EFRAG & GRI landmark Statement of Cooperation*. European Financial Reporting Advisory Group, & Global Reporting Initiative. www.efrag.org

EFRAG. (2021b). *Final Report, PROPOSALS FOR A RELEVANT AND DYNAMIC EU SUSTAINABILITY REPORTING STANDARD-SETTING*. https://www.efrag.org/Lab2#subtitle2

Ellerman, D. (2011). Double-Entry Accounting: The Mathematical Formulation and

Electronic copy available at: https://ssrn.com/abstract=4336278

Generalization. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.1340619

Ellerman, D. P. (1986). Double entry multidimensional accounting. *Omega*, *14*(1), 13–22.

European Commission. (2021). How Does the EU Taxonomy Fit Within the Sustainable Finance Framework? In *European Commission*. https://ec.europa.eu/info/sites/default/files/business_economy_euro/banking_and_finance/documents/sustainable-finance-taxonomy-factsheet_en.pdf

Feger, C., & Mermet, L. (2017). A blueprint towards accounting for the management of ecosystems. *Accounting, Auditing and Accountability Journal*, *30*(7), 1511–1536. https://doi.org/10.1108/AAAJ-12-2015-2360

Feger, C., & Rambaud, A. (2020). *Improving nature's visibility in financial accounting.: Method 3- Comprehensive accounting with respect to ecology (CARE) model.* https://hal.science/hal-02976915/document

Fellingham, J. C. (2007). Is Accounting an Academic Discipline? *Accounting Horizons*, *21*(2). https://doi.org/10.2308/acch.2007.21.2.159

Ferraro, F., Etzion, D., & Gehman, J. (2015). Tackling Grand Challenges Pragmatically: Robust Action Revisited. *Organization Studies*, *36*(3), 363–390. https://doi.org/10.1177/0170840614563742

Frame, B., & O'Connor, M. (2011). Integrating valuation and deliberation: The purposes of sustainability assessment. *Environmental Science and Policy*, *14*(1), 1–10. https://doi.org/10.1016/j.envsci.2010.10.009

Freeman, M. C., & Groom, B. (2013). Biodiversity valuation and the discount rate problem. *Accounting, Auditing and Accountability Journal*, *26*(5), 715–745. https://doi.org/10.1108/AAAJ-02-2013-1226

Garst, J., Maas, K., & Suijs, J. (2022). Materiality Assessment Is an Art, Not a Science: Selecting ESG Topics for Sustainability Reports . *California Management Review*, *2022*(1), 000812562211206. https://doi.org/10.1177/00081256221120692/ASSET/IMAGES/LARGE/10.1177_00081256221120692-FIG2.JPEG

Electronic copy available at: https://ssrn.com/abstract=4336278

GRI 3: Material Topics 2021, GRI Standards 1 (2022). https://www.globalreporting.org/umbraco/Surface/ResourceCentre/PopupResource?id=12453

GRI - Sector Program. (n.d.). Retrieved December 6, 2022, from https://www.globalreporting.org/standards/sector-program/

Guenther, E., Endrikat, J., & Guenther, T. W. (2016). Environmental management control systems: a conceptualization and a review of the empirical evidence. *Journal of Cleaner Production*, *136*, 147–171. https://doi.org/10.1016/j.jclepro.2016.02.043

Hopwood, A. G., Unerman, J., & Fries, J. T. A.-T. T.-. (2010). *Accounting for sustainability: practical insights*. Earthscan.

Houdet, J., Ding, H., Quétier, F., Addison, P., & Deshmukh, P. (2020). Adapting double-entry bookkeeping to renewable natural capital: An application to corporate net biodiversity impact accounting and disclosure. *Ecosystem Services*, *45*, 101104. https://doi.org/10.1016/J.ECOSER.2020.101104

Humphrey, C., O'Dwyer, B., & Unerman, J. (2017). Re-theorizing the configuration of organizational fields: the IIRC and the pursuit of 'Enlightened' corporate reporting. *Accounting and Business Research*, *47*(1), 30–63. https://doi.org/10.1080/00014788.2016.1198683

IAASB. (2021). *Handbook of international quality control, auditing, review, other assurance, and related services pronouncements.* International Federation of Accountants. https://www.ifac.org/iaasb/publications/2020-handbook-international-quality-control-auditing-review-other-assurance-and-related-services

IFRS Foundation. (2023). *IFRS Foundation welcomes culmination of TCFD work and transfer of TCFD monitoring responsibilities to ISSB from 2024*. https://www.ifrs.org/news-and-events/news/2023/07/foundation-welcomes-tcfd-responsibilities-from-2024/

IFRS Foundation, & Global Reporting Initiative. (2023). *GRI establishes Sustainability Innovation Lab in coordination with the IFRS Foundation*. https://www.ifrs.org/news-and-events/news/2023/11/gri-establishes-sustainability-innovation-lab-in-coordination-with-the-ifrs-foundation/

Electronic copy available at: https://ssrn.com/abstract=4336278

Ijiri, Y. (1967). *The foundations of accounting measurement: A mathematical, economic, and behavioral inquiry*. Prentice-Hall.

Ijiri, Y. (1975). *Theory of accounting measurement* (Issue 10). American Accounting Association.

Jia, J., Ranger, N., & Chaudhury, A. (2022). Designing for comparability: A foundational principle of analysis missing in carbon reporting systems. *SSRN Electronic Journal*. https://doi.org/10.2139/SSRN.4258460

Jones, M. J. (2010). Accounting for the environment: Towards a theoretical perspective for environmental accounting and reporting. *Accounting Forum*, *34*(2), 123–138. https://doi.org/10.1016/j.accfor.2010.03.001

Kaplan, R. S., & Norton, D. P. (2001a). Transforming the Balanced Scorecard from Performance Measurement to Strategic Management: Part I. *Accounting Horizons*, *15*(1), 87–104. https://doi.org/10.2308/ACCH.2001.15.1.87

Kaplan, R. S., & Norton, D. P. (2001b). Transforming the Balanced Scorecard from Performance Measurement to Strategic Management: Part II. *Accounting Horizons*, *15*(2), 147–160. https://doi.org/10.2308/ACCH.2001.15.2.147

Kaplan, R. S., & Ramanna, K. (2021). How to Fix ESG Reporting. *SSRN Electronic Journal*. https://doi.org/10.2139/SSRN.3900146

KPMG. (2020). *The time has come - The KPMG Survey of Sustainability Reporting 2020*.

Maas, K. (2018). Do Corporate Social Performance Targets in Executive Compensation Contribute to Corporate Social Performance? *Journal of Business Ethics*, *148*(3), 573–585. https://doi.org/10.1007/s10551-015-2975-8

Marrone, M., Linnenluecke, M. K., Richardson, G., & Smith, T. (2020). Trends in environmental accounting research within and outside of the accounting discipline. *Accounting, Auditing and Accountability Journal*, *33*(8), 2167–2193. https://doi.org/10.1108/AAAJ-03-2020-4457

Mattessich, R. (1964). *Accounting and analytical methods: measurement and projection of income and wealth in the micro-and macro-economy*. Scholars Book Company.

Electronic copy available at: https://ssrn.com/abstract=4336278

O'Dwyer, B. (2011). The Case of Sustainability Assurance: Constructing a New Assurance Service. *Contemporary Accounting Research*, *28*(4), 1230–1266. https://doi.org/10.1111/j.1911-3846.2011.01108.x

O'Dwyer, B., & Unerman, J. (2016). Fostering rigour in accounting for social sustainability. *Accounting, Organizations and Society*, *49*, 32–40. https://doi.org/10.1016/j.aos.2015.11.003

Peters, G. P. (2010). Carbon footprints and embodied carbon at multiple scales. *Current Opinion in Environmental Sustainability*, *2*(4), 245–250. https://doi.org/10.1016/J.COSUST.2010.05.004

Rajgopal, S. (2021). Integrating practice into accounting research. *Management Science*, *67*(9), 5430–5454. https://doi.org/10.1287/MNSC.2020.3590

Rambaud, S. C. (2010). *Algebraic models for accounting systems*. World Scientific.

Rennert, K., Prest, B. C., Pizer, W. A., Newell, R. G., Anthoff, D., Kingdon, C., Rennels, L., Cooke, R., Raftery, A. E., Ševč\'\iková, H., & others. (2021). The Social Cost of Carbon: Advances in Long-Term Probabilistic Projections of Population, GDP, Emissions, and Discount Rates. *Brookings Papers on Economic Activity*.

Ring, I., Hansjürgens, B., Elmqvist, T., Wittmer, H., & Sukhdev, P. (2010). Challenges in framing the economics of ecosystems and biodiversity: the TEEB initiative. *Current Opinion in Environmental Sustainability*, *2*(1–2), 15–26.

Science, K. M.-B., & 2023, undefined. (n.d.). Kunming-Montreal global biodiversity framework: an important global agenda for biodiversity conservation. *Biodiversity-Science.NetK MaBiodiversity Science, 2023•biodiversity-Science.Net*. Retrieved May 21, 2024, from https://www.biodiversity-science.net/EN/article/downloadArticleFile.do?attachType=PDF&id=97648

Spencer, A. W., & Webb, T. Z. (2015). Leases: A Review of Contemporary Academic Literature Relating to Lessees. *Accounting Horizons*, *29*(4), 997–1023. https://doi.org/10.2308/ACCH-51239

Stoelhorst, J. W., & Vishwanathan, P. (2024). Beyond Primacy: A Stakeholder Theory of Corporate Governance. *Academy of Management Review*, *49*(1), 107–134.

Electronic copy available at: https://ssrn.com/abstract=4336278

https://doi.org/10.5465/amr.2020.0268

Sukhdev, P. (2009). Costing the Earth. *Nature 2009 462:7271*, *462*(7271), 277–277. https://doi.org/10.1038/462277a

Sustainability reporting standards interim draft - EFRAG. (n.d.). Retrieved December 6, 2022, from https://www.efrag.org/Activities/2105191406363055/Sustainability-reporting-standards-interim-draft#

Taskforce on Nature-related Financial Disclosures, & S. B. T. N. (2023). *Guidance for corporates on science-based targets for nature*.

ten Have, O. (1973). *De geschiedenis van het boekhouden*. Delwel.

The Value Reporting Foundation. (2021). *IIRC and SASB form the Value Reporting Foundation, providing comprehensive suite of tools to assess, manage and communicate value*.

Thornton, D. B. (2013). Green accounting and green eyeshades twenty years later. *Critical Perspectives on Accounting*, *24*(6), 438–442. https://doi.org/10.1016/J.CPA.2013.02.004

Uhrynuk, M., Burdulia, A., & Lee, J. C. Y. (2021). *China Proposes New Environmental And Social Disclosures For Listed Companies*. Mondaq.Com & Mayer Brown.

Unerman, J. (2020). Risks from self-referential peer review echo chambers developing in research fields: 2018 Keynote Address presented at The British Accounting Review 50th Anniversary Celebrations, British Accounting and Finance Association Annual Conference, London. *The British Accounting Review*, *52*(5), 100910. https://doi.org/10.1016/J.BAR.2020.100910

Unerman, J., Bebbington, J., & O'Dwyer, B. (2018). Corporate reporting and accounting for externalities. *Accounting and Business Research*, *48*(5), 497–522. https://doi.org/10.1080/00014788.2018.1470155

Directive - 2023/970, (2023). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv%3AOJ.L_.2023.132.01.0021.01.ENG&toc=OJ%3AL%3A2023%3A132%3ATOC

UNRISD. (2023). *Authentic sustainability assessment: A user manual for the sustainable development performance indicators (SDPI)*.

Electronic copy available at: https://ssrn.com/abstract=4336278

Velte, P. (2020). Do CEO incentives and characteristics influence corporate social responsibility (CSR) and vice versa? A literature review. *Social Responsibility Journal*, *16*(8), 1293–1323. https://doi.org/10.1108/SRJ-04-2019-0145

Williams, D. (1991). Probability with Martingales. In *Probability with Martingales*. Cambridge University Press. https://doi.org/10.1017/CBO9780511813658

World Economic Forum. (2023). *World Economic Forum and ISSB partner to compile learnings on early sustainability reporting efforts*. https://www.weforum.org/press/2023/06/world-economic-forum-and-issb-partner-to-compile-learnings-on-early-sustainability-reporting-efforts/

Yamey, B. S. (1964). Accounting and the Rise of Capitalism: Further Notes on a Theme by Sombart. *Journal of Accounting Research*, *2*(2), 136. https://doi.org/10.2307/2489995

Yamey, B. S. (2008). Double-entry bookkeeping. In S. N. Durlauf & L. E. Blume (Eds.), *The New Palgrave Dictionary of Economics: 2nd edition* (pp. 1432–1435). Palgrave macmillan.

Yu, K. (2013). Does Recognition versus Disclosure Affect Value Relevance? Evidence from Pension Accounting. *The Accounting Review*, *88*(3), 1095–1127. https://doi.org/10.2308/ACCR-50381

Zappettini, F., & Unerman, J. (2016). 'Mixing' and 'Bending': The recontextualisation of discourses of sustainability in integrated reporting. *Discourse and Communication*, *10*(5), 521–542. https://doi.org/10.1177/1750481316659175

Electronic copy available at: https://ssrn.com/abstract=4336278

A. Appendices

A.1 Mathematical appendix

Equivalence of record-keeping systems Proposition 2

To prove Proposition 2, it is without loss of generality to show a bijection between two of these systems. First, define $\kappa^l = \{\frac{1}{\lambda_1^l}, \dots, \frac{1}{\lambda_n^l}\}$, this is a vector holding the multiplicative inverse values of the elements in λ^l. This vector must exist, as we required that none of the elements in the vector λ are equal to zero. Now take some list $A \in \mathbb{R}^n$ with elements $\alpha_i = \lambda_i q_i$ recording the state of each element in the organization and its corresponding value. If A satisfies Axiom 5* for $\lambda = \mathbf{1}$ (i.e. all prices are 1). Then we know that $\sum_{i=1}^{n} \alpha_i = \sum_{i=1}^{n} 1 \cdot q_i = 0$, and thus $A \in W_{\mathbf{1}}$. We then define the transformation:

$$A \mapsto^{\kappa^l} \{\kappa^l a_1, \dots, \kappa^l a_n\} = k^l(A)$$

We can directly verify that:

$$\lambda^l k^l(A) = \sum_{i=1}^{n} \lambda_i^l \kappa_i^l a_i = \sum_{i=1}^{n} 1\, a_i = 0$$

This shows that $k^l(A) \in W_{\lambda^l}$. Furthermore, every k^ldefines a linear mapping from R^n to R^n that is invertible, one-to-one, and covers the entire space. Hence, it defines a bijection for every element in A. Which means k^l defines a bijective mapping $W_1 \mapsto^{\kappa^l} W_{\lambda^l}$, while the inverse of this mapping is determined in the same way through λ^l. As we have bijective mappings from any W_{λ^l} to and from W_1, to find a bijective mapping from W_{λ^l} to W_{λ^m}, simply combine the two relevant mappings:

$$W_{\lambda^l} \mapsto^{\kappa^l} W_1 \mapsto^{\lambda^k} W_{\lambda^k}$$

The isomorphism of Figure 1 shows how to relate any of these systems to the debit-credit notation and completes the result.

Electronic copy available at: https://ssrn.com/abstract=4336278

A.2 Bookkeeping in this mathematical notation

At this point, we have not discussed any of the normal accounting objects like debits, credits, ledgers, or journal entries. To show how they relate to the list of n values that forms an accounting statement in this section, we go through two steps. First, we show how a list of positive and negative values can be translated to the normal debit and credit notation (and back) via a one-to-one mapping. Second, we give an example of what posting a journal entry to the books would look like in this framework.

A bookkeeping account, like a ledger account, can be modelled mathematically as an ordered tuple with two non-negative entries, one debit value, and one credit value. We will denote such tuples as elements $\alpha_i = [d_i//c_i]$ where $d_i > 0$ is the debit value held in account i and $c_i > 0$ the credit value held in this account. To go from an element α to a value on the real line, like the values defined by the q_i, one takes the difference between the debit and credit value in the given account to get the net-debit value: $val(\alpha_i) = d_i - c_i$. This definition of value also works the other way round, as any real number r > 0 (r < 0) can be mapped to an equivalence class of elements $val^{-1}(r) = [r + x//x]$ as long as x, r + x > 0. This reverse mapping thus implies that two accounts have the same value and are thus equivalent in terms of value if the difference between their debit and credit entries is the same.

Under these definitions, we have a one-to-one mapping from the bookkeeping accounts to the real line, shown in Figure 1. This mapping is not accidental. Double-entry bookkeeping developed before the use of negative numbers took root. This construction of an account where positive credit values are used to denote negative debit values allows one to work with negative quantities without specifying negative numbers. In that way, bookkeepers could work with events with negative value (reductions in value) before the negative numbers to express them were invented.

Electronic copy available at: https://ssrn.com/abstract=4336278

Once we use the mapping from the real line to the debit-credit notation, the list of values generated by the q_i becomes a list of α-accounts that can be used to construct ledgers, balance sheets, and journal entries. To do so, we join n α-accounts into an ordered list.

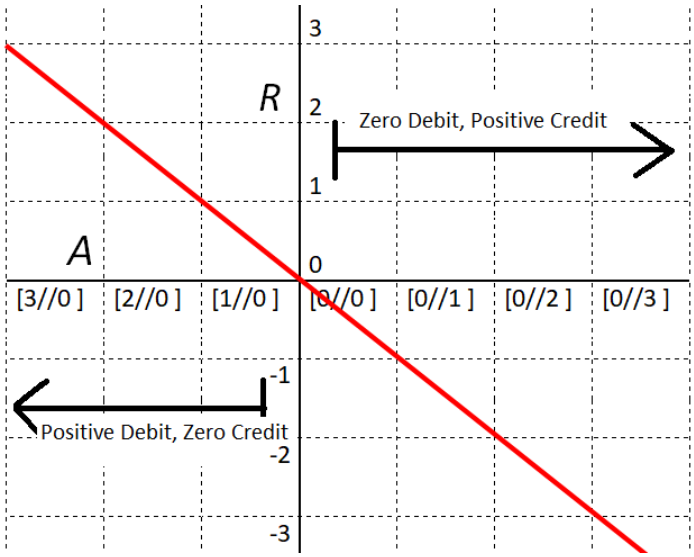


Figure 3 One-to-one mapping from the real values to debit-credit notation

Each account i holds the value specified by the corresponding measure q_i(after the mapping $val^{-1}(q_i)$is applied). The result is a list of accounts that keep track of the values in the organization in debit-credit form. Figure 2 shows a prototypical bookkeeping system and makes the links between the mathematical entities and the corresponding bookkeeping entities clearer.

Both the ledger in Panel A and the journal entry in Panel B of Figure 2, have the properties implied by our axioms. Both are ordered lists of accounts each with a unique name. In every account in the list, the value is found by taking the difference between the debit and the credit value of the account. For the accounts on the balance sheet, this difference corresponds to $q_i(\omega^\tau)$ the measure for the entire history of the company from incorporation till the balance date τ. For a

Electronic copy available at: https://ssrn.com/abstract=4336278

journal entry, the value in each account equals the value of the measure for the economic event ω_τ.

a_1 [x_1 // 0]	a_1 [0 // y_1]	a_1 [x_1 // y_1]
a_2 [x_2 // 0]	a_2 [y_2 // 0]	a_2 [$x_2 + y_2$ // 0]
a_3 [x_3 // 0]	a_3 [0 // 0]	a_3 [x_3 // 0]
a_4 [0 // x_4]	a_4 [0 // 0]	a_4 [0 // x_4]
a_5 [0 // x_5]	a_5 [0 // 0]	a_5 [0 // x_5]
a. Opening ledger	b. Journal entry	c. Updated ledger

Index, *i*	Account name
1	Cash
2	Materials Inventory
3	Completed goods Inventory
4	Owner's equity
5	Loan, 5%
...	

d. Translation table from indexes to account names

Figure 4 Processing a journal entry in the mathematical framework

We process the journal entry in Panel B by adjusting the accounts in the original ledger in Panel A with the indicated amounts. Since adding 0 to a real number does not affect the value we can (and do) ignore zero-valued accounts in a given journal entry, and only show the accounts whose value is affected. After the journal entry has been processed, the updated ledger in Panel C is again an ordered collection of α-accounts, like the original ledger and the journal entry, so that processing the journal entry does not change the nature of a ledger. Via the indexes and their translation to real-world objects shown in Panel D, we can see that the journal entry in Panel B credits cash and debits materials, so corresponds to a cash purchase of materials.

Electronic copy available at: https://ssrn.com/abstract=4336278

A.3 Regional differences

In the main text, we highlighted industry differences by plotting the Social Cost of Carbon against profit per industry, here we make a similar figure but split our observations based on the continents the companies are headquartered in. No clear patterns are showing between geographic regions, but this is likely because most of the data is from the Americas. In the rest of the analysis, we, therefore, drop this distinction.



Figure 5, Regional patterns in performance measures

A.4 Time differences

So far, we have looked at the value flows as more-or-less consistent entities over time. Our data covers 10 years and there is good reason to believe that there are time trends in the data. The focus on CO2 reduction has increased markedly over the last decades and profit swings up and down with the economic cycle. The table just below shows the number of observations, company-years, per year. It shows that we have sufficient observations in our data in all 10

Electronic copy available at: https://ssrn.com/abstract=4336278

years. CO2 and tax-adjusted performance is the simple sum of the profits, taxes, and value destroyed through CO2 emissions. To visualize each of these value flows and their development over the ten years in our data. We see a mild downward trend in the cost of carbon emissions, which is consistent with a decrease in emissions by the reporting companies. Given that these companies choose to measure and report their carbon emissions, it is likely that they are also attempting to reduce these emissions. Profit and taxes appear to co-move, which is to be expected as the main determinant of corporate income taxes is the income in the current year. While there is no clear trend in profits, taxes seem to go down over time, which is consistent with (Dyreng et al., 2017)

Table 1, Observations per year

2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
311	371	451	533	537	558	559	489	438	429	424

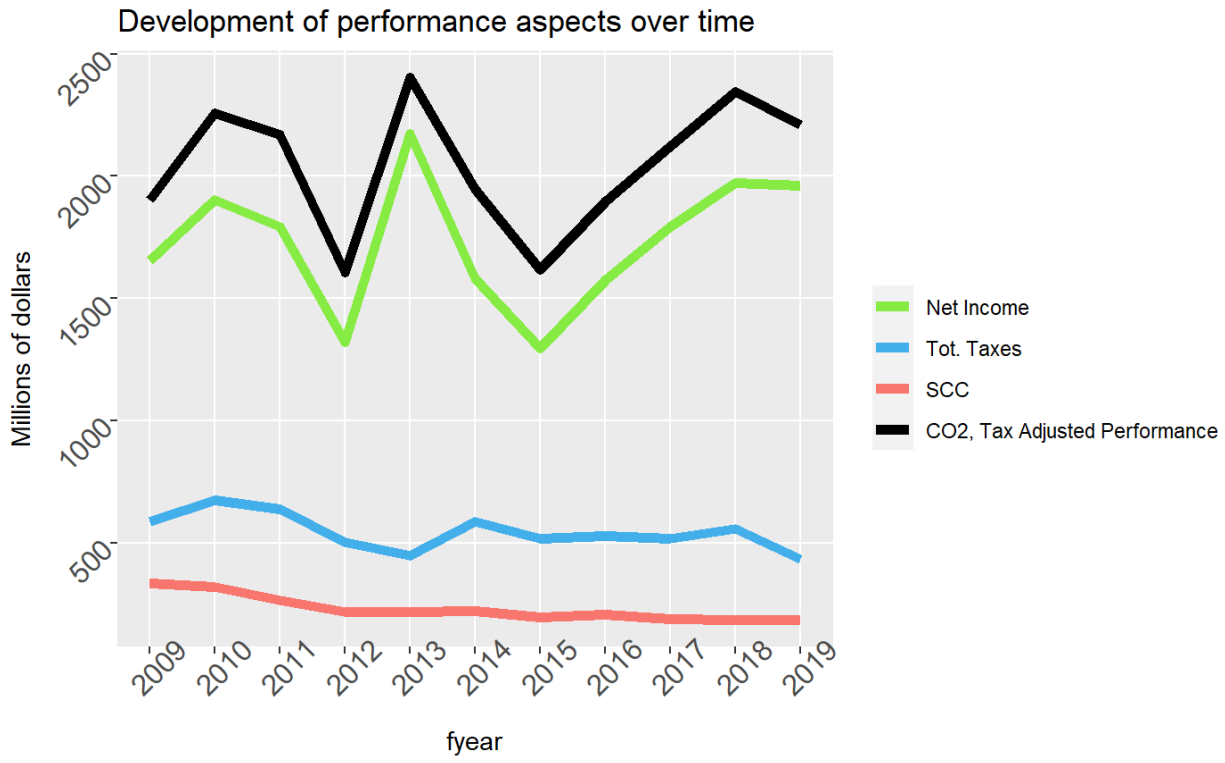


Figure 6 Development of the three performance measures over time

Electronic copy available at: https://ssrn.com/abstract=4336278